Supplement 5: Apparatus

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What it is, and why it is here rather than in the main text

The examples reported in Part III were located with the help of an apparatus built to search for them automatically. That apparatus is a tool. It embodies no claim about MetaCAs, and its results are about the search rather than about the substrate, so it is documented here in full and referenced from the main text only where the provenance of an example matters.

It is documented in full for one reason: it did not do what it was built to do, and the specific ways in which it failed are reusable. The intended object was a system that adjusts its own operating point using only quantities it can compute at runtime, toward the coexistence of frozen and moving structure that the main text calls the third sense of the edge of chaos. What exists is a system that scores a parameter surface, moves over it under a learned surrogate, and reports where it went. The step from the second to the first was not taken, and the closing subsection states precisely which part of it is missing.

The two coordinates

The apparatus varies two parameters of the selection rule that chooses among candidate updates at a site.

Policy precision β grades how sharply a policy is chosen from its expected free energy: at $\beta \to 0$ selection is uniform over candidates, and as β grows it concentrates on the score-minimising candidate. We write it β throughout this supplement to avoid collision with the causal-currency gain γ of *The coupling as a dial*, which is a different quantity; the implementation calls it `policy-precision`.

Epistemic weight κ scales the information-seeking term of that score against the goal-directed term. At $\kappa = 0$ the rule is purely exploitative.

Both are swept geometrically: $\beta \in \{1, 2, 4, 8, 16, 32, 64\}$ and $\kappa \in \{0, 0.1, 0.2, 0.5, 1\}$, giving thirty-five cells, each run at four seeds.

The objective, and its validation

The apparatus scores a run by an intrinsic measure rather than by damage reach, because damage reach is computed on a twin run and is therefore unavailable to the system itself. The phenotype spacetime sheet is tiled into 100×100 patches, each packed contiguously to a bitstring and compressed; the observable is the *fraction of patches whose compression ratio lies in* $[0.3, 0.9]$, which distinguishes a graded distribution from a bimodal one.

Patch size and packing are both load-bearing and both produced defects worth recording. At $S = 50$ the compressor's own header dominates a 312-byte payload and the incompressible end of the scale saturates, so several distinct regimes score identically at the ceiling. And packing patch rows to whole bytes appends four zero bits per row, a period-13 regularity present in every patch that the compressor exploits; the first version of the validation table was wrong for this reason, and correcting it removed an apparent winner.

Validated against elementary rules of undisputed regime, the corrected measure separates in the intended direction: rules 54 and 110 score 94% and 75%, while 204 (frozen), 90 (nested) and 30 (chaotic) all score 0%. It is worth noting that it places rule 90 with rule 30, declining to call it class IV despite its fractal appearance, where damage reach places rule 90 next to the frozen rule.

The surface, and the ridge

Table 1 gives the objective over the thirty-five cells.

The surface has two arms and a valley between them. One arm runs along low β and survives every value of κ; the other occupies high β at $\kappa \leq 0.1$ and vanishes abruptly at $\kappa = 0.2$. Between them, at $\beta = 4$ to 16, the objective falls to a minimum of 0.109 and in three cells to zero. Approached from below the ridge is a broad plateau; approached from above it is a cliff.

This geometry is the useful product of the search, and it is not what the search was built to produce. Two facts about it are established in the main text and should be read

Table 1

The objective over the search space, averaged over four seeds. Nonzero values occupy an L-shaped ridge: low β at any κ, and any β at $\kappa \leq 0.1$. Everything else is identically zero.

$\beta \backslash \kappa$	0.0	0.1	0.2	0.5	1.0
1	0.594	**0.641**	0.562	0.562	0.625
2	0.422	0.422	0.406	0.438	0.469
4	0.312	0.219	0.109	0.250	0.000
8	0.000	0.453	0.156	0.000	0.000
16	0.203	0.484	0.047	0.000	0.000
32	0.609	**0.625**	0.000	0.000	0.000
64	0.484	0.547	0.000	0.000	0.000

together with the table. The high-β arm consists of configurations that reach an absorbing state — $\beta = 32$ at $\kappa = 0.1$ freezes permanently at $t = 1046$ — while the low-β arm does not. And the examples exhibiting criticality lie in the valley, at $\beta = 8$ to 10, scoring *below* both arms. The objective's two maxima are a living regime and a dying one, and it ranks the dying one highest.

The search paths

Figure 1 shows twelve-episode trajectories from four starting points under a surrogate-driven policy, drawn twice: over the control space the apparatus manipulates, and over the two-dimensional observable space in which it perceives its own position.

The right-hand panel carries the substantive limitation. Of 595 cell pairs, five collide within 0.02 in observable space — most consequentially $(\beta = 2, \kappa = 0.5)$ with $(\beta = 8, \kappa = 0.1)$, which the main text shows to be dynamically very different. A policy reading only these two observables is blind along that ridge. This is documented rather than repaired.

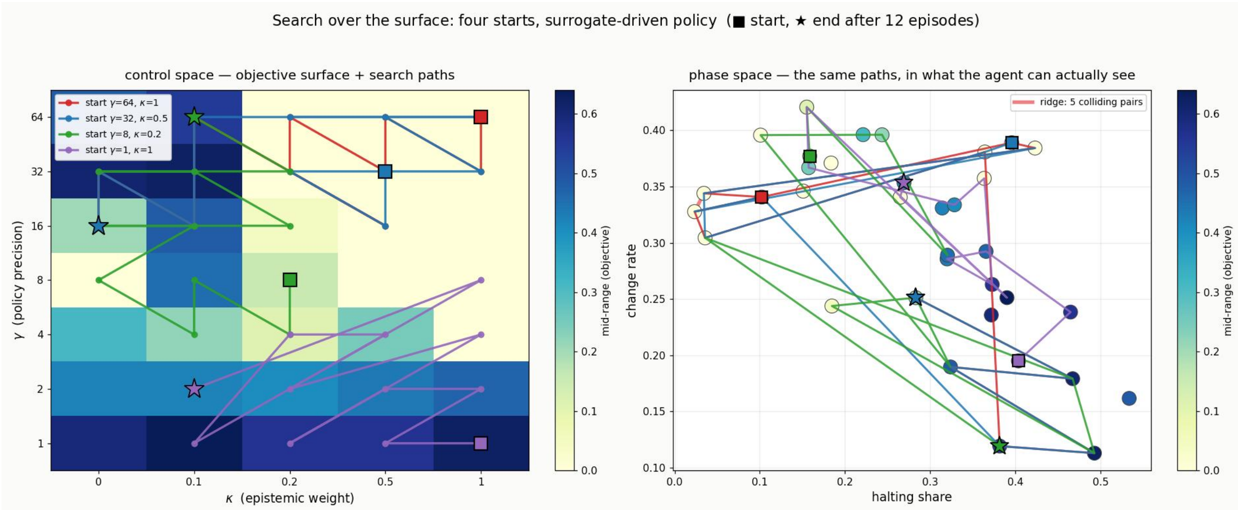

Figure 1

Search over the surface. Left: the objective with four search paths in control space (κ, β)*. Right: the same paths in the observable space the apparatus can actually see — halting share against change rate. The marked ridge joins the five cell pairs that collide within* 0.02 *in that space; along it the apparatus cannot distinguish its own position.*

The screen that should have come first

An earlier controller in this family targeted a different pair: a knob λ weighting a conatus term, with realised winner-hunger as its sensor. Sweeping λ across its entire range moves mean realised hunger by 0.005, against a seed-level spread an order of magnitude larger. A closed-loop sweep over ten target values then found every target settling at a boundary, with a bifurcation between 0.160 and 0.170: below it λ runs to one, above it to zero, and nothing rests between. The critical target was not an attractor but the mean realised hunger itself, separating positive error forever from negative error forever. A sign controller on a flat plant has no interior fixed point.

That controller could not have worked, and this was establishable before it was built. The screen we now apply to any candidate pair (u, y) has four conditions, each cheap: does u move y on-policy by more than seed noise (*leverage*); does the target lie inside y's attainable range (*reachability*); does y track the objective rather than merely being movable (*validity*); and does the closed loop have a fixed point rather than a repeller (*stability*). The λ–hunger pair fails leverage, fails reachability twice over, and fails stability. Validity was never reached. Applied to the other knobs, the screen selects β, which passes the first three outright — and β had not been treated as a control at all.

The bridge test

The obstruction that then appears is different in kind from the one λ met, and it is the reason the loop remains open. Damage reach is measured by forking a run, perturbing one bit, advancing a twin and counting divergences. No system can compute it about itself. So a controller on β cannot use the sensor that makes β attractive, and the question becomes whether any runtime-computable quantity tracks a target worth reaching.

Two leave-one-out tests answer it, and they answer it differently:

predictors → target	held-out R^2
internal score → damage reach (external, twin-run)	+0.016
(halting share, change rate) → local compressibility (intrinsic)	+0.726

Same system, same class of observable, same form of test; the difference is whether the target is something the system can perceive. The estimator was smoke-tested on synthetic signal (+0.976) and synthetic noise (−0.165) before being run on real data, and the objective column was reproduced identically across two independently written implementations, so the contrast is attributable to the system rather than to the analysis. An earlier value of +0.541 for the second row reflected a window mismatch between the observable and objective runs; measuring both on the matched window recovers +0.726.

The conclusion is narrow and worth stating exactly. **The loop is closable — just not on damage reach.** A pair of quantities the system computes anyway predicts an intrinsic objective well. That result survives the discovery, reported in the main text, that the objective itself ranks configurations against their long-run survival, because it is a statement about coupling between sensor and target rather than about the value of the target.

What was not achieved

No self-driving search was built. The apparatus scores a surface and moves over it; it does not adjust its own operating point toward criticality using only runtime quantities, and it was never run in a closed loop against a live system. The examples in Part III were found by this apparatus together with our own reading of what it returned — our choice of which cells to run long, our noticing that the highest-scoring cells were the ones that died — and we cannot separate the two contributions. Where the main text reports an example, it reports it as an object with measured properties, not as the output of an autonomous procedure.

Three further gaps are worth recording against any attempt to close the loop. The objective, as the main text establishes, is not the right target: it rewards the frozen phase that precedes permanent freezing, and a controller ascending it would ascend toward death. The observable space is degenerate along the ridge of Figure 1, so even a correct target would be pursued blind in that region. And the stability condition of the screen was never tested for β, because the target it would have been tested against did not survive. A controller on this apparatus therefore remains unbuilt for a reason that is now specific rather than general: not that the system cannot be steered, but that we do not yet have a target worth steering toward that it can also see.

A second apparatus, for an experiment not run here

The preceding sections describe a search over an existing construction. This one describes how a *selection* experiment over the same substrate should be specified before it is built. The two share a discipline and not a subject: there the question was whether a knob could move its sensor, here it is whether a population could exhibit the effect the experiment is built to detect. The material is retained because the conditions below are checkable in advance, and because the experiment they govern — adding a selective filter, so that plasticity alters what selection sees — is the extension the main text names and does not carry out.

Specifying a Selection Experiment Before Running It

A selection experiment reports trajectories. Nothing in a trajectory says whether the experiment was capable of exhibiting the effect it was built to detect, and a run that could not have succeeded produces the same shape of output as one that tried and failed. The apparatus described here exists to separate those two cases in advance rather than after the fact.

The target as a certificate

We state the target as an object that must be exhibited, not as a pattern to be recognised in a plot. A genome carries a rule at each locus together with a flag marking that locus as held; a held locus keeps its inherited rule for the whole evaluation. Writing dep for the dependence of a genome on within-evaluation rewriting and perf for its raw function, a completed assimilation claim is a path $g_0, \ldots, g_n$ satisfying, simultaneously:

1. every step $g_i \to g_{i+1}$ is admitted by the declared genetic operators;
2. $\mathrm{perf}(g_i) \geq \tau$ for every i, so the path never leaves the functional region;
3. $\mathrm{dep}(g_{i+1}) \leq \mathrm{dep}(g_i)$ at every step, and $\mathrm{dep}(g_n) < \mathrm{dep}(g_0)$ strictly;
4. g_n is static — every locus held; and
5. perf of g_n with every locus held still clears τ.

The last condition is the one most easily omitted. A population can reduce its measured dependence without its inherited field being able to carry the behaviour alone, and only an explicit held-rule evaluation distinguishes the two.

Selection, however, does not rank raw function. It ranks fitness $\mathrm{fit}_c = \mathrm{perf} - c\,\mathrm{dep}$. A path can therefore satisfy all five conditions and still be untraversable, because a step that preserves function may lose fitness. Adding the requirement $\mathrm{fit}_c(g_{i+1}) \geq \mathrm{fit}_c(g_i)$ gives a strictly stronger object, and makes a conflict explicit that is otherwise rediscovered empirically: a dependence-reducing step keeps its fitness exactly when

$$\mathrm{perf}(g_i) - \mathrm{perf}(g_{i+1}) \;\le\; c\Big(\mathrm{dep}(g_i) - \mathrm{dep}(g_{i+1})\Big), \tag{1}$$

that is, when the function a hold costs is no more than the cost it saves. If the inequality fails at every accessible dependence-reducing step, no traversable path exists at any length, and (1) identifies c as the parameter that would dissolve the obstruction.

Conditions checkable before the run

Two consequences are checkable without running anything. First, if dependence takes a single value across the reachable genomes, the strict-decrease condition cannot be met and no certificate of any length exists; a constant axis is therefore a disqualifying property of the design, not a null result. Second, and less obviously, an axis on which the score is non-constant may still be untraversable: a profile that is zero at every level but one has two distinct values and a single transition, and a hill-climber cannot cross a lone cliff. Counting the adjacent transitions that carry gradient, rather than counting distinct values, is what distinguishes a navigable axis from a spike.

This distinction is not hypothetical here. Under the gain parameterisation of the main text's endogenous gate, sampled at eight uniform levels, the band score is zero at seven of them and positive at the eighth. The design is disqualified by the second criterion while passing the first, and re-resolving the parameter where the gain curve is convex is what makes the axis traversable — a change to the experiment rather than to the check.

Invariants the implementation must satisfy

The specification above constrains the target. A second family of conditions constrains the apparatus, and each is stated as an executable predicate tested against the specific failure that motivated it, since a check written from the description of an invariant rather than from its counterexample can pass the defect it was introduced to catch.

Tape alignment. Both branches of a perturbation fork must consume identical random draws whatever their gates decide. A gate that reads the phenotype fires at different loci in the two branches, so a closed gate that skips a draw desynchronises the tapes and the divergence that follows is an artefact of the generator rather than a causal effect. The discipline is to draw unconditionally and then decide how to use the draw; the check must also confirm the branches actually differed, since equal draw counts are trivially equal when they do not.

Layering fidelity. Any extension must reproduce the measurement it extends when its additional machinery is in the neutral position. Determinism does not imply fidelity: an extension can reproduce byte-for-byte and still report a different value than the construction it claims to generalise.

Linked inheritance. Every heritable component must travel together under recombination. Transferring a rule without its held flag places it under an unrelated hold pattern, so the operator no longer tests the mechanism it was introduced for.

Operator reachability. Every value of every gene must be attainable under mutation. A value the operator cannot reach is present in the representation and absent from the experiment.

Empirical null. Any claim of the form "rarer than chance" requires a no-selection control running the identical lifecycle. An analytical baseline must match the lifecycle actually implemented: with elitist retention, in which survivors are not re-mutated, the mutation-only expectation for a symmetric flip rate μ after n generations is $\frac{1}{2}\Big(1-(1-\mu)^n\Big)$ rather than the familiar $\frac{1}{2}\Big(1-(1-2\mu)^n\Big)$.

Treatment separation. Selection ranks scores, so a term that shifts every genome equally cannot act. Arms differing in such a term will coincide, and their coincidence is a diagnostic of an inert treatment rather than a robust result.

Two of these deserve emphasis because they invert a natural reading. The empirical null is not a formality: in the runs reported here, dependence falls and the held fraction rises under no selection at all, purely because holds accumulate by mutation. Declining dependence is therefore not evidence of assimilation, and a treatment must be compared against that trajectory rather than against zero. Treatment separation is the complementary case: identical arms indicate that the parameter never reached selection.

Diagnosing a negative result

When a run produces no certificate, the specification does not say why, and the three possible reasons call for different remedies. Holding each locus of a high-performing plastic genome at each admissible rule separates them. If no rule preserves function at a locus, the plastic computation there is not representable by any single static rule. If some rule preserves function but the inherited allele does not, the representation suffices and the difficulty is that mutation cannot prepare the allele before it is fixed. If the inherited allele itself preserves function and improves fitness, an assimilation step was available and the search did not take it. Only the first is a statement about the substrate.

The distinction is decidable cheaply. Testing each locus at its current allele costs one evaluation per locus rather than one per locus and rule, and settles the third case outright when it fires.

What the apparatus does not establish

These conditions are necessary, not sufficient. They constrain what an experiment can claim; they do not make its measurements correct, and no invariant here would detect a mis-specified dynamics that satisfies all of them. They are also specific to the construction studied in Part III: the certificate assumes a genome with per-locus holding and a scalar cost, and neither the conditions nor the diagnostic transfers unchanged to a different architecture. Finally, the calibration used by the score must share dynamics with the constructions placed on it; boundary conditions that differ between a calibration and the objects it calibrates can reorder the reference rules themselves, and where that holds the resulting band assignments are not comparable.